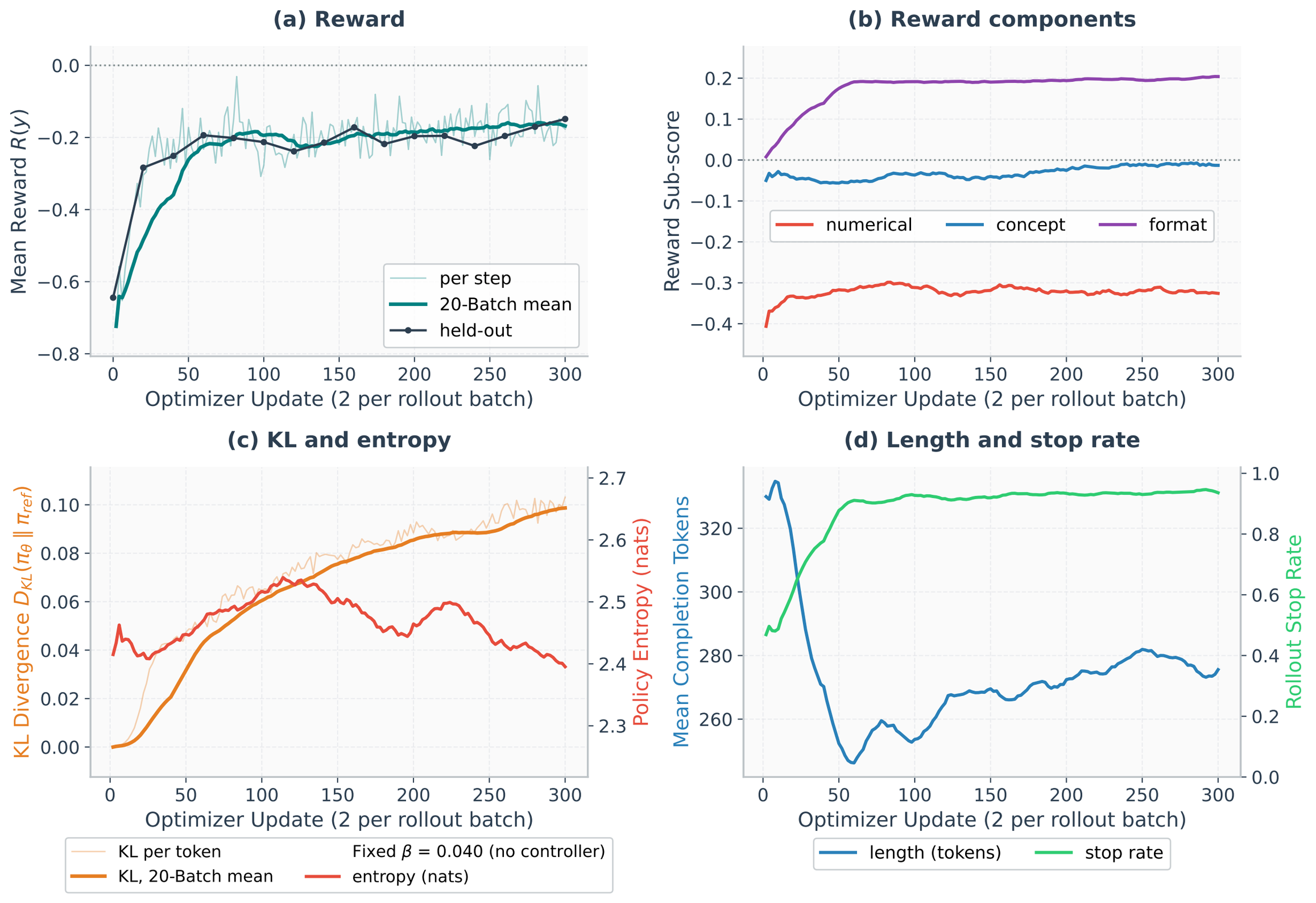
(a) Reward
Mean Reward $R(y)$
0.0
−0.2
−0.4
−0.6
−0.8
per step
20-Batch mean
held-out
0
50
100
150
200
250
300
Optimizer Update (2 per rollout batch)
(b) Reward components
Reward Sub-score
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
numerical
concept
format
Optimizer Update (2 per rollout batch)
(c) KL and entropy
KL Divergence $D_{KL}(\pi_\theta \| \pi_{ref})$
0.10
0.08
0.06
0.04
0.02
0.00
Policy Entropy (nats)
2.7
2.6
2.5
2.4
2.3
Optimizer Update (2 per rollout batch)
KL per token
KL, 20-Batch mean
Fixed β = 0.040 (no controller)
entropy (nats)
(d) Length and stop rate
Mean Completion Tokens
320
300
280
260
Rollout Stop Rate
1.0
0.8
0.6
0.4
0.2
0.0
Optimizer Update (2 per rollout batch)
length (tokens)
stop rate